31. When you look at an object very close to your eyes, the : **1**

(A) Ciliary muscles of your eye contract and the eye lens becomes thick.

(B) Ciliary muscles of your eye get relaxed and the eye lens becomes thick.

(C) Ciliary muscles of your eye contract and the eye lens becomes thin.

(D) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

CBSE 26

(a) How does the focal length of eye lens change as the distance of the object from the eyes is altered ? Explain. CBSE 26 2

(a) How does the focal length of eye lens change as the distance of the object from the eyes is altered ? Explain. CBSE 26 *2*

(a) • If the distance of the object from the eye is increased, ciliary muscles relax, lens becomes thin and so the focal length increases. • If the distance between the object from the eye is decreased, the ciliary muscles contract, lens becomes thick and the focal length decreases.	**1** **1**

30. The pair of eye parts responsible for admitting different amount of light in to the eyes is **1**

(A) Iris and pupil

(B) Ciliary muscles and pupil

(C) Retina and Iris

(D) Ciliary muscles and cornea

30. The pair of eye parts responsible for admitting different amount of light in to the eyes is **1**

(A) Iris and pupil

(B) Ciliary muscles and pupil

(C) Retina and Iris

(D) Ciliary muscles and cornea

CBSE 26

(A) / Iris and pupil

34. (A) (i) How does the change in curvature of the eye lens helps us in the process of seeing the nearby objects clearly ?

(ii) State the range of the power of accommodation of a normal human eye. CBSE 26 2

34. (A) (i) How does the change in curvature of the eye lens helps us in the process of seeing the nearby objects clearly ?

(ii) State the range of the power of accommodation of a normal human eye. CBSE 26 **2**

(A) (i) While looking at objects nearer to eye, the curvature of the eye lens increases. Consequently, the focal length of the eye lens decreases.	**1**
(ii) From 25cm (near point) to infinity (far point)	**1**

31. (a) State one important function of the following parts of the human eye :

(i) Retina

(ii) Pupil

(b) State the role of ciliary muscles in focussing objects at varying distances from the eye. *3*

31. (a) State one important function of the following parts of the human eye :

(i) Retina

(ii) Pupil

(b) State the role of ciliary muscles in focussing objects at varying distances from the eye. *3*

(a) (i) Retina behaves like a light sensitive screen on which image of an object is formed.	1
(ii) Pupil : regulates and controls the amount of light entering the eye.	1
(b) When ciliary muscles relax or contract they change the curvature of eye lens and hence the focal length to focus objects at varying distances from the eye. **[Alternate answer of (b):** It regulates the focal length or thickness of the lens.**]**	1

(b) Write the main function of each of the following in the human eye :

(i) Cornea

(ii) Iris

(iii) Retina

(iv) Ciliary muscles *2*

(b) Write the main function of each of the following in the human eye :

(i) Cornea

(ii) Iris

(iii) Retina

(iv) Ciliary muscles *2*

(b)	
(i) Cornea – Most of the refraction of light rays falling on the eye.	½
(ii) Iris – controls the size of the pupil.	½
(iii) Retina – image is formed on it	½
(iv) Ciliary muscles – To modify the curvature / focal length of the eye lens.	½

12. The change in the focal length of an eye lens in human beings is caused by the action of **1**

(a) optic nerves (b) ciliary muscles

(c) retina (d) cornea

12. The change in the focal length of an eye lens in human beings is caused by the action of 1

(a) optic nerves (b) ciliary muscles

(c) retina (d) cornea

(b) ciliary muscles

11. In human eye the part which allows light to enter into the eye is – **1**

(a) Retina

(b) Pupil

(c) Eye lens

(d) Cornea

11. In human eye the part which allows light to enter into the eye is – **1**

(a) Retina (b) Pupil

(c) Eye lens (d) Cornea

(d) Cornea

13. The lens system of human eye forms an image on a light sensitive screen, which is called as : **1**

(a) Cornea

(b) Ciliary muscles

(c) Optic nerves

(d) Retina

13. The lens system of human eye forms an image on a light sensitive screen, which is called as : 1

(a) Cornea

(b) Ciliary muscles

(c) Optic nerves

(d) Retina

(d) /Retina

31. (a) Define the term power of accommodation of human eye. Write the name of the part of eye which plays a major role in the process of accommodation and explain what happens when human eye focuses (i) nearby objects and (ii) distant objects. **3**

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(a)	
• Ability of the eye lens to adjust its focal length.	1
• Ciliary muscles	1
• (i) While focusing on nearby objects ciliary muscles contract, eye lens becomes thick and its focal length decreases.	½
(ii) While focusing on distant objects ciliary muscles relax, eye lens becomes thin and its focal length increases.	½

13. Consider the following statements in the context of human eye : **1**

(a) The diameter of the eye ball is about 2.3 cm.

(b) Iris is a dark muscular diaphragm that controls the size of the pupil.

(c) Most of the refraction for the light rays entering the eye occurs at the crystalline lens.

(d) While focusing on the objects at different distances the distance between the crystalline lens and the retina is adjusted by ciliary muscles.

The correct statements are –

(A) (a) and (b) (B) (a), (b) and (c)

(C) (b), (c) and (d) (D) (a), (c) and (d)

13. Consider the following statements in the context of human eye : **1**

(a) The diameter of the eye ball is about 2.3 cm.

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The correct statements are –

(A) (a) and (b) (B) (a), (b) and (c)

(C) (b), (c) and (d) (D) (a), (c) and (d)

(A)/ (a) and (b)

27. Define the term power of accommodation of human eye. What happens to the image distance in the eye when we increase the distance of an object from the eye ? Name and explain the role of the part of human eye responsible for it in this case. *3*

27. Define the term power of accommodation of human eye. What happens to the image distance in the eye when we increase the distance of an object from the eye ? Name and explain the role of the part of human eye responsible for it in this case. *3*

•Ability of the eye lens to adjust its focal length.	1
• Image distance remains unchanged	1
•Ciliary muscles –	½
While focusing on distant objects ciliary muscles relax, eye lens becomes thin and its focal length increases.	½

(b) Name the muscles present in the human eye which enable it to focus on objects at varied distances (i.e., distant as well as nearby objects). Explain how it happens. *2*

(b) Name the muscles present in the human eye which enable it to focus on objects at varied distances (i.e., distant as well as nearby objects). Explain how it happens. *2*

(b) • Ciliary muscles • Ciliary muscles adjust/change the focal length of eye lens by changing the curvature of eye lens.	1 1

18. **Assertion (A) :** When ciliary muscles contract, eye lens becomes thin.

Reason (R) : Ciliary muscles control the power of the eye lens. 1

18. **Assertion (A) :** When ciliary muscles contract, eye lens becomes thin.

Reason (R) : Ciliary muscles control the power of the eye lens. 1

(d) / Assertion (A) is false but Reason (R) is true.

13. The curvature of eye lens of human eye **1**

(A) is fixed.

(B) can be increased.

(C) can be decreased.

(D) increases or decreases as the case may be.

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(A) is fixed.

(B) can be increased.

(C) can be decreased.

(D) increases or decreases as the case may be.

(D) / increases or decreases as the case may be

14. Most of the refraction for the light rays entering the eye occurs at 1

(A) Iris (B) Pupil

(C) Crystalline lens (D) Outer surface of Cornea

14. Most of the refraction for the light rays entering the eye occurs at **1**

(A) Iris (B) Pupil

(C) Crystalline lens (D) Outer surface of Cornea

(D) / Outer surface of cornea

14. The part of human eye which controls the amount of light entering into it. **1**

(A) Iris (B) Cornea

(C) Ciliary muscles (D) Pupil

14. The part of human eye which controls the amount of light entering into it. **1**

(A) Iris (B) Cornea

(C) Ciliary muscles (D) Pupil

(D) Pupil

14. When we focus our eyes on a distance object, the ciliary muscles of our eyes are relaxed. In this case the eye lens becomes
    (A) thin and its curvature is maximum.
    (B) thin and its curvature is minimum.
    (C) thick and its curvature is minimum.
    (D) thick and its curvature is maximum.

14. When we focus our eyes on a distance object, the ciliary muscles of our eyes are relaxed. In this case the eye lens becomes

(A) thin and its curvature is maximum.

(B) thin and its curvature is minimum.

(C) thick and its curvature is minimum.

(D) thick and its curvature is maximum.

(B)/ thin and its curvature is minimum

13. In the human eye, most of the refraction for the light rays entering the eye occurs at the surface of :

(A) Cornea

(B) Crystalline lens

(C) Pupil

(D) Aqueous humor

13. In the human eye, most of the refraction for the light rays entering the eye occurs at the surface of :

(A) Cornea

(B) Crystalline lens

(C) Pupil

(D) Aqueous humor

(A) / Cornea

13. The part of the human eye which can modify the curvature of the eye lens to some extent is :

(A) Pupil

(B) Cornea

(C) Ciliary muscles

(D) Aqueous humour

13. The part of the human eye which can modify the curvature of the eye lens to some extent is :

(A) Pupil

(B) Cornea

(C) Ciliary muscles

(D) Aqueous humour

(C)/ Ciliary muscles

5. The image distance from the eye lens in the normal eye when we increase the distance of an object from the eye *1*

(A) increases.

(B) decreases.

(C) remains unchanged.

(D) depends on the size of the eyeball.

5. The image distance from the eye lens in the normal eye when we increase the distance of an object from the eye *1*

(A) increases.

(B) decreases.

(C) remains unchanged.

(D) depends on the size of the eyeball.

(C) / remains unchanged

Person suffering from cataract has

(a) elongated eyeball

(b) excessive curvature of eye lens

(c) weakened ciliary muscles

(d) opaque eye lens **1**

Person suffering from cataract has

(a) elongated eyeball

(b) excessive curvature of eye lens

(c) weakened ciliary muscles

(d) opaque eye lens 1

$\boxed{\text{(d) opaque eye lens}}$

Define the term power of accommodation. Write the modification in the curvature of the eye lens which enables us to see the nearby objects clearly ?

Define the term power of accommodation. Write the modification in the curvature of the eye lens which enables us to see the nearby objects clearly ?

• Power of accommodation – Ability of eye lens to adjust its focal length.	1
• Curvature increases/lens becomes thick	1

Trace the sequence of events which occur when a bright light is focused on your eyes.

Trace the sequence of events which occur when a bright light is focused on your eyes.

Receptor Cells of eyes/retina → Sensory Neuron → Brain / CNS ↓ Pupil contracts / Eye lids close/blink ← Eye Muscles ← Motor Neuron (Note: If a child writes spinal cord in place of brain give full credit to him/her)	½x6

What happens to the image distance in the normal human eye when we decrease the distance of an object, say 10 m to 1 m ? Justify your answer.

What happens to the image distance in the normal human eye when we decrease the distance of an object, say 10 m to 1 m ? Justify your answer.

• Image distance remains the same. • It is the distance between the eye lens and retina, which remains the same.	1 1

Write the structure of eye lens and state the role of ciliary muscles in the human eye.

Write the structure of eye lens and state the role of ciliary muscles in the human eye.

Structure – Fibrous, jelly like structure	1
Role – To change the curvature of eye lens / to change the focal length of eye lens.	1

(a) Write the function of each of the following parts of human eye :

(i) Cornea (ii) Iris (iii) Crystalline lens (iv) Ciliary muscles

(b) Why does the sun appear reddish early in the morning ? Will this phenomenon be observed by an astronaut on the Moon ? Give reason to justify your answer.

(a) Write the function of each of the following parts of human eye :

(i) Cornea (ii) Iris (iii) Crystalline lens (iv) Ciliary muscles

(b) Why does the sun appear reddish early in the morning ? Will this phenomenon be observed by an astronaut on the Moon ? Give reason to justify your answer.

a. Function of: • Cornea: focuses light rays / permits the light to enter the eye.. • Iris: Controls amount of light entering the eye. / controls the size of pupil. • Crystalline Lens: Converges light rays onto retina. • Cilliary Muscles: Adjusts focal length of eye lens by contraction and relaxation so that sharp image can be obtained on the retina. / helps in accommodation	½ x 4
b. In early morning, sun light has to cover larger distance in the atmosphere. So, the shorter wavelengths scatter out. Only the longer wavelengths like red reach our eye.	1 ½
On moon – No	½
Cause: Moon has no atmosphere	1

(a) Write the function of each of the following parts of human eye :

cornea; iris; crystalline lens; ciliary muscles

(a) Write the function of each of the following parts of human eye :

cornea; iris; crystalline lens; ciliary muscles

• Cornea – Refracts the rays of light falling on the eye	½
• Iris – Controls the size of the pupil	½
• Crystalline lens – Focuses the image of the object on the retina	½
• Ciliary muscles – Holds the eye lens and adjusts its focal length	½